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Introduction

This guide explores how selected Artificial Intelligence (AI) tools can support learners with neurodevelopmental disorders, with a focus on the study of Modern Languages. As digital technologies become increasingly integrated into education, AI offers new opportunities to reduce barriers to learning, promote independence, and provide personalised support tailored to individual needs. For students with conditions such as dyslexia, autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD), and other neurodevelopmental differences, AI tools can help make language learning more accessible and engaging.

Modern Language assessments often require students to demonstrate a broad range of skills, including reading, writing, listening, speaking, grammar, vocabulary acquisition, and cultural understanding. These tasks can present additional challenges for neurodivergent learners, particularly where difficulties with processing, organisation, memory, concentration, or communication are involved. AI technologies can provide practical assistance by offering real-time feedback, personalised learning pathways, speech-to-text and text-to-speech capabilities, translation support, vocabulary reinforcement, and study organisation tools.

The intended audience for this guide are undergraduate students at Cardiff University, with a focus on those studying Modern Languages. This guide covers a wide range of functionality provided by selected AI tools, how to use them, and how they can help Modern Languages students with task preparation. This guide is a baseline for how AI tools can be used, and in no way covers every single use of every single tool covered, just ways in which we have identified could help students. This guide provides a baseline overview of how AI tools may be used to support neurodivergent students. It does not seek to cover every possible application of the tools included but rather highlights selected uses that were identified as potentially beneficial across different aspects of the student experience. The development of the guide was further informed by feedback from volunteer students within the Modern Languages (MLANG) community, whose reflections and lived experiences helped refine its content, relevance, and accessibility.

AI tools should be viewed as supplementary support mechanisms rather than replacements for existing reasonable adjustments, specialist support services, or inclusive teaching practices. The effectiveness of any tool will vary according to individual needs, preferences, and contexts, and students should be encouraged to identify the approaches that best support their learning.

Why these disorders?

What is neurodiversity

Neurodiversity is the concept of that there is natural variation in the ways that the human brain works, signifying that there is no one “correct” way to think, learn, or process information.

“Neurodiversity is the concept that there is natural variation in how people’s brains work, with no single “correct” way, and that a wide range of perceptions and responses to the world should be accepted and encouraged” – (Child Mind Institute, 2026).

What disorders are we covering, and why?

Dyslexia

Evidence

Dyslexia effects reading accuracy, spelling, phonological processing and vocabulary. These difficulties can impact reading comprehension, grammar, and written language production in Modern Languages. (University of Strathclyde, n.d.)

Rationale for inclusion

Modern Language relies heavily on reading, writing, spelling, and pronunciation. AI tools that can perform text-to-speech, speech recognition, and grammar support that can provide huge benefits for those who struggle.

Possible link to assessments/ tasks

- Writing exams
- Reading exams
- Vocabulary
- Pronunciation activities

Dyspraxia

Evidence

Dyspraxia can affect motor coordination, planning and organisation, handwriting speed and notetaking. People with dyspraxia may also struggle with planning (e.g. creating a plan for tasks and projects). (Dyspraxia Foundation (2012)

Rationale for inclusion

Written tasks and language learning activities often require extensive notetaking and written production, AI based speech-to-text may reduce these barriers. AI tools can also be used to effectively plan tasks and create timelines.

Possible link to assessment/tasks

- Notetaking
- Written Tasks
- Project Planning

Attention Deficit Hyperactivity Disorder (ADHD)

Evidence

ADHD can affect concentration, working memory, time management and task completion. Learners may struggle to remember vocabulary, organise revision, or sustain attention span during lengthy lessons and tasks. (Royal Borough of Windsor and Maidenhead, n.d.)

Rationale for inclusion

Modern Languages tasks require sustained focus and structured responses. AI tools can support planning, revision planning, resource organisation, and personalised learning.

Possible links to assessments/ tasks

- Writing task
- Listening tasks
- Revision and coursework management

Autism (autism spectrum disorder)

Evidence

Autistic learners may experience challenges with social communication, interpreting figurative language, or understanding contextual clues. Language learning often involves cultural interpretation and communication skills. (Reachout ASC (n.d.)).

Rationale for inclusion

Modern Languages requires both linguistic and social communication competencies. AI tools can provide structured explanations, conversational practise and personalised learning environments.

Possible links to assessments/ tasks

- Speaking assessments
- Cultural interpretation tasks
- Group tasks

Processing disorders

Evidence

Processing disorders can affect auditory, visual, and speed of processing. This means that tasks such as reading, listening, following instructions, vocabulary retention, and language comprehension may be challenging, particularly in the study of Modern Languages. (National Center for Learning Disabilities (n.d.))

Rationale for inclusion

Modern Language Learning requires learners to quickly process spoken language and written information, remember new vocabulary, and interpret meaning from context. AI tools that provide text-to-speech, content summarisation, and personalised pacing can help reduce barriers for learners with processing difficulties

Possible link to assessments/ tasks

- Writing exams
- Reading exams
- Vocabulary
- Multi step instructions
- Note taking

Why is a recourse like this relevant to Cardiff University, and education as a whole?

Inclusive education and recognition of learners at Cardiff University with neurodevelopmental disorders is very important. Neurodivergence is real and impacts people in all different areas of life. Cardiff University has 7% (1,980) of students that have disclosed that they are neurodivergent as of the 25/26 academic year, along with 7.8% (40) of MLANG students that have also disclosed the fact they are neurodivergent.

With this number of students that have disclosed that they are neurodivergent, it is still very common for people to get to the point of being in the first year of their university course, unaware that they are neurodivergent, and therefore do not get the support they need. Promoting this topic is important, as it will allow for people to get support they did not know they needed, helping further contribute to equality in education.

Statistics from higher education state that neurodiversity is increasingly important in the sector, with recent statistics showing that 22% of UK students have a neurodivergent diagnosis, with up to 28% identifying as neurodivergent in some way. This contributes to an overrepresentation of neurodivergence among students compared to the general population, of which only 15-20% are believed to be neurodivergent (Eves, 2026). This highlights the growing importance of neurodiversity in higher education and reinforces the need for targeted strategies and resources that enable neurodivergent students to thrive academically.

Why these tools?

We conducted a comprehensive review of a wide range of tools to identify those most suitable for neurodivergent students at Cardiff University. The evaluation began with an initial shortlist of 11 tools, which were assessed against a set of rigorous criteria to determine their suitability and effectiveness. The criteria used in the evaluation are outlined below:

Evaluation criteria and rating of selected AI tools

What we are evaluating these tools for:

Criterion	Definition	Indicators	Rating (1-5)
Accessibility	The extent to which this tool is accessible for the targeted students with neurodevelopmental disorders	Text-to-speech, clear language, keyboard accessibility, simple interface, adjustable text size, captions.	1 = Poor 5 = Outstanding
Personalisation	The features provided which allow the users to change the app, to promote a more user specific experience	Tailored explanations, adjustable difficulty, personalised feedback, response customisation.	1 = Poor 5 = Outstanding
Cognitive load and management	The extent that the tool can help reduce unnecessary mental effort	Step-by-step guidance, concise responses, chunking of information, minimal distractions.	1 = Poor 5 = Outstanding
Pedagogical Usefulness	How well the AI tool supports meaningful learning and problem solving.	Accurate explanations, scaffolding, constructive feedback, encourages critical thinking.	1 = Poor 5 = Outstanding

The criteria for scoring:

Score	Meaning
1	Does not meet the criteria, not fit or purpose
2	Meets the criteria to a limited extent, improvements would be needed
3	Meets the criteria moderately, basic use possible
4	Meets the criteria, intended use very possible
5	Meets the criteria fully, intended use is excellently provided.

First, we started creating a series of relevant categories, aligned to features in an app that would be useful to our target audience. We then created a criterion for scoring across characteristics of every category we have defined.

Below are the scoring criteria:

Accessibility:

Indicator	What to look for	Rating 0 – 5
Clear and readable interface	Clean layout, readable fonts, minimal clutter	
Accessible communication	Uses plain language and can simplify explanations	
Multiple ways to interact	Supports text, voice, images, or other input/output methods	
Computability with accessibility features	Works with screen readers, keyboard navigation, text-to-speech, etc.	
Sensory accessibility	Avoids overwhelming visuals or unnecessary distractions	
Total	Points average	

Personalisation:

Indicator	What to look for	Points 0 – 5
Adaptive explanations	can simplify, expand, or rephrase content	
Adjustable response style	User can request different formats, lengths, or levels of detail	
Personalised feedback	Feedback reflects the learner's responses rather than generic advice	
Supports individual learning pace	Learners can revisit, repeat, or build on previous explanations	
Flexibility	Can generate examples, quizzes, summaries, or alternative explanations	
Total	Points average	

Cognitive load management

Indicator	What to look for	Points 0 – 5
Chunking information	Breaks information into manageable sections	
Step by step guidance	Explains tasks sequentially	
Concise explanations	Avoids unnecessary complexity or information overload	
Organisation	Uses headings, bullet points, summaries, or logical structure	
Attention support	Keeps responses focused and avoids irrelevant information	
Total	Points	

Pedagogical usefulness

Indicator	What to look for	Points 0 – 5
Concept explanation	Explains reasoning rather than only giving answers	
Constructive feedback	Identifies mistakes and explains how to improve	
Encourages critical thinking	Prompts learners to think rather than simply copy	
Supports independent learning	Helps learners solve problems themselves	
Learning reinforcement	Provides examples, practice questions, summaries, or revision support	
Total	Points	

From this point, we made note of an average score from a section, and add it to every other average score to get a total score for the app. From this we were able to see how every tool we analysed compared with each other, and which tools aligned best with the needs of students with neurodevelopmental disorders.

Below is the selection of tools we analysed, along with the task they were best aligned for:

Reading	Writing	Listening	Speaking
• Perplexity • Notebook LM • Knowt • Language tool • Wolfram alpha • Photomath	• Microsoft Copilot (w/ voice) • Quillbot	• Google Gemini (w/ live mode)	• Google Gemini (w/ live mode)

Below is the result from our analysis:

Final scores:

AI tool	Score
Perplexity	16.4 / 20
Notebook LM	15.6 / 20
Knowt	15.6 / 20
Quillbot	16.5 / 20
Language tool	12.9 / 20
Wolfram alpha	10.8 / 20
Photo math	11.8 / 20
Microsoft Copilot (w/ voice)	16.2 / 20
Google Gemini (w/ live mode)	16.5 / 20

Score breakdown:

Score range	Description
0 – 8.9	Not fit for purpose
9 – 11.9	Fit for purpose but may have limited use case.
12 – 13.9	Fit for purpose
14 – 15.9	Fit for purpose, has great features, and a good use case
16 or more	Fit for purpose, has excellent features, and a great use case.

From this we were able to create a further shortlist of tools:

Cardiff University Provide Gemini Notebook, Google Gemini, and Microsoft Copilot. This also helped us when deciding which tools to shortlist.

The final shortlisted tools:

Reading	Writing	Listening and speaking
• Perplexity • Notebook LM • Language tool	• Microsoft Copilot (w/ voice) • Quillbot	• Google Gemini (w/ live mode)

How did Google Gemini Perform in our analysis?

Google Gemini (w/ voice)

Accessibility:

Indicator	What to look for	Rating 0 – 5
Clear and readable interface	Clean layout, readable fonts, minimal clutter	4
Accessible communication	Uses plain language and can simplify explanations	4
Multiple ways to interact	supports text, voice, images, or other input/output methods	5
Computability with accessibility features	Works with screen readers, keyboard navigation, text-to-speech, etc.	5
Sensory accessibility	Avoids overwhelming visuals or unnecessary distractions	5
Total		Points average: 4.6

Personalisation:

Indicator	What to look for	Points 0 – 5
Adaptive explanations	can simplify, expand, or rephrase content	4
Adjustable response style	User can request different formats, lengths, or levels of detail	4
Personalised feedback	Feedback reflects the learner's responses rather than generic advice	4
Supports individual learning pace	Learners can revisit, repeat, or build on previous explanations	4
Flexibility	Can generate examples, quizzes, summaries, or alternative explanations	5
Total		Points average: 4.2

Cognitive load management:

Indicator	What to look for	Points 0 – 5
Chunking information	Breaks information into manageable sections	4
Step by step guidance	Explains tasks sequentially	4
Concise explanations	Avoids unnecessary complexity or information overload	4
Organisation	Uses headings, bullet points, summaries, or logical structure	4
Attention support	Keeps responses focused and avoids irrelevant information	4
Total		Points average: 4

Pedagogical usefulness:

Indicator	What to look for	Points 0 – 5
Concept explanation	Explains reasoning rather than only giving answers	4
Constructive feedback	Identifies mistakes and explains how to improve	4
Encourages critical thinking	Prompts learners to think rather than simply copy	4
Supports independent learning	Helps learners solve problems themselves	4
Learning reinforcement	Provides examples, practice questions, summaries, or revision support	4
Total		Points average: 4

SCORE: 16.8 / 20

Learner profiles

Every student experience learning differently. To illustrate how neurodiversity can shape academic experiences, we've created a series of learner profiles based on common student experiences that learners that are neurodivergent may have. You can find these learner profiles below:

Profile 1: Hyperactive ADHD (Adam)

- **Demographics:** 18, 1st Year BA Modern Languages (English native, learning Spanish).
- **Core Profile:** Outgoing and highly creative, but struggles with executive dysfunction, time management, hyperfixation shifts, and finishing multi-step tasks.

Profile Dimension	Details & Strategies
Strengths	High energy, creativity, hyperfocus capacity, enthusiasm, strong social/conversational potential.
Academic Friction	Initiating unstructured study; managing long-term essay deadlines; remaining focused during long grammar lectures.
AI Matching Strategy	Use AI to externalise executive function and gamify micro-tasks.
Concrete AI Workflows	• Microsoft Copilot: "Break down my 2,000-word Spanish literature essay into 5 daily micro-tasks. Put deadlines on each and format as a checklist for Microsoft To-Do." • LanguageTool: Passively catches grammar/spelling errors while drafting without requiring dedicated editing sessions. • Perplexity AI: Focuses research queries to avoid falling down irrelevant internet "rabbit holes."
Assessment-Targeted Tasks	• Targeted Task (Coursework Essays): Prompt Copilot to build a structured 7-day writing roadmap with micro-deadlines. • Targeted Task (Grammar & Syntax Exams): Use LanguageTool for immediate feedback loops while practicing past papers. • Targeted Task (Literature & Culture Projects): Use Perplexity to gather verified, in-scope academic sources quickly without distraction.
Pedagogical Guardrail	Set strict time limits on AI interactions so hyperfocus isn't spent tweaking AI prompts instead of studying.

Profile 2: Inattentive ADHD (Amelia)

- **Demographics:** 18, 1st Year BA Modern Languages (English native, learning French).
- **Core Profile:** Deep thinker with strong pattern recognition but paralysed by high activation energy (task initiation), low executive resilience, and anxiety around foreign language speaking.

Profile Dimension	Details & Strategies
Strengths	Deep thinking, empathy, curiosity, conceptual pattern recognition.
Academic Friction	High task paralysis when staring at blank pages; anxiety surrounding real-time oral speaking in front of peers.
AI Matching Strategy	Use AI to lower initiation friction and provide low-stakes, judgement-free practice.
Concrete AI Workflows	• Google Gemini (Live Mode) / Copilot Voice: Low-stakes practice holding basic French conversations without peer-judgment or performance anxiety. • NotebookLM: Upload module reading PDFs to generate 3-bullet point summaries, reducing cognitive load before diving into full texts.
Assessment-Targeted Tasks	• Targeted Task (Oral Speaking Exams): Practice holding low-stakes, conversational roleplays with Gemini Live Mode to lower anxiety prior to real oral exams. • Targeted Task (Reading Comprehension Exams): Use NotebookLM to generate initial core summaries of dense set texts before reading the full material. • Targeted Task (Essay Planning): Prompt Copilot to generate 3 different starter outlines to overcome blank-page initiation paralysis.
Pedagogical Guardrail	Ensure she uses conversational AI as a stepping stone to build confidence for real human interaction, not as a permanent substitute.

Student Guide | Google Gemini

Profile 3: Dyslexia (Samuel)

- **Demographics:** 19, 1st Year BA Modern Languages (Welsh national, limited Welsh, passionate about new languages).
- **Core Profile:** Resilient and creative, but experiences severe working memory load and processing friction when decoding written text, symbols, and spelling.

Profile Dimension	Details & Strategies
Strengths	Resilience, verbal communication, creative problem solving.
Academic Friction	Reading dense foreign text; accurate spelling/grammar in target languages; decoding written exam materials.
AI Matching Strategy	Transition from text-heavy interaction to multimodal (audio/visual) learning .
Concrete AI Workflows	• QuillBot & LanguageTool: Dual-proofreading layer to catch subtle grammatical inflections and orthographic errors. • Google Gemini (Multimodal Vision): Snap a photo of a printed Spanish text or worksheet → request audio read-back and simplified translation breakdowns. • NotebookLM: Convert dense syllabus PDFs into automatically generated audio overviews/podcasts and practice quizzes.
Assessment-Targeted Tasks	• Targeted Task (Written Composition & Reports): Use QuillBot and LanguageTool as a final proofreading step to catch spelling and grammar slips. • Targeted Task (Translation & Text Analysis): Snap photos of physical texts using Gemini to hear them read aloud while reviewing syntax. • Targeted Task (Revision & Knowledge Checks): Convert lecture slides and notes into automated practice quizzes using NotebookLM.
Pedagogical Guardrail	Require double-checking QuillBot output against class grammar rules, as AI suggestions can occasionally alter target-language nuances.

Profile 5: Slow Processing Speed (Declan)

- **Demographics:** 18, 1st Year BA Modern Languages (Bilingual Welsh/English).
- **Core Profile:** Accurate and deep thinker when given time but struggles to keep up with fast-paced live lectures and rapid spoken audio.

Profile Dimension	Details & Strategies
Strengths	High accuracy, deep conceptual understanding, methodical work habits.
Academic Friction	Taking real-time notes during lectures; rapid oral comprehension exercises; fast-paced seminar discussions.
AI Matching Strategy	Use AI for asynchronous pacing control and information distillation .
Concrete AI Workflows	• Copilot / Gemini Voice: Upload audio/video recordings of lectures to generate full transcripts, allowing Declan to review at his own pace. • NotebookLM: "Summarise this lecture transcript into core key takeaways and a 10-word vocabulary list to reduce reading volume."
Assessment-Targeted Tasks	• Targeted Task (Listening Comprehension Tests): Generate transcripts of practice audio using Copilot/Gemini to study sentence structure asynchronously. • Targeted Task (End-of-Term Exam Preparation): Use NotebookLM to condense voluminous term readings into high-priority core summary sheets. • Targeted Task (Seminar Preparation): Use NotebookLM to create pre-class key vocabulary lists to reduce real-time cognitive processing load.
Pedagogical Guardrail	Ensure audio transcripts are reviewed alongside lecture slides to verify technical term accuracy.

Profile 4: Autistic Learner (Austin)

- **Demographics:** 18, 1st Year BSc Computer Science (Welsh national, English speaking).
- **Core Profile:** Highly analytical with deep attention to detail, but vulnerable to sensory overload, unpredictable environment shifts, and figurative/ambiguous language.

Profile Dimension	Details & Strategies
Strengths	Logical thinking, detail accuracy, systematic problem solving, high reliability.
Academic Friction	Interpreting non-literal assignment prompts; managing sensory overload during crowded lectures; navigating group work.
AI Matching Strategy	Use AI as a clarifier to strip away ambiguity and create explicit structure .
Concrete AI Workflows	• Copilot / NotebookLM: Direct prompt: "Rewrite these assignment instructions into literal, step-by-step instructions. Remove metaphorical language and list explicit deliverables." • NotebookLM: Converts lecture notes into structured visual diagrams, explicit glossaries, and factual Q&As to study in a quiet, low-sensory setting.
Assessment-Targeted Tasks	• Targeted Task (Interpreting Assignment Briefs): Paste module rubrics into Copilot to generate explicit, step-by-step deliverable checklists. • Targeted Task (Theoretical/Conceptual Exams): Use NotebookLM to compile literal, factual glossaries and visual diagrams from course files. • Targeted Task (Group Project Planning): Use Copilot to clearly divide tasks and write unambiguous communication templates for team members.
Pedagogical Guardrail	Explicitly define parameters for AI use so Austin understands exact academic integrity boundary lines.

Profile 6: Auditory Processing Disorder - APD (Sebastian)

- **Demographics:** 18, 1st Year BA Modern Languages (English native, limited Spanish).
- **Core Profile:** Visual learner who struggles to distinguish and process spoken auditory input, leading to fatigue in listening exercises.

Profile Dimension	Details & Strategies
Strengths	Visual processing capacity, spatial learning, resilience, hands-on learning.
Academic Friction	Foreign language listening exams; auditory lectures; background noise in oral seminars.
AI Matching Strategy	Convert auditory input into visual text and visual diagrams .
Concrete AI Workflows	• Google Gemini / Copilot Live: Live transcription of spoken class audio into formatted, real-time written text. • NotebookLM: Input audio recordings to produce written study summaries, mind maps, and flashcard sets, completely bypassing auditory fatigue.
Assessment-Targeted Tasks	• Targeted Task (Audio/Listening Assessments): Practice with Gemini by pairing spoken audio outputs directly with real-time screen transcription. • Targeted Task (Lecture Content Mastery): Process lecture audio through NotebookLM to create visual mind maps and text notes. • Targeted Task (Vocabulary Building): Convert audio flashcards into visual/written vocabulary matrices.
Pedagogical Guardrail	Target language phonology still needs practice; use synchronised text-and-audio playback tools rather than cutting out listening entirely.

Disclaimers

- Always use your Cardiff University email address to access any AI tools recommended by the University. Where available, use the University-provided or University-approved versions of these tools. This helps ensure that your data is protected under Cardiff University's security, privacy and data protection arrangements.
- Avoid sharing personal, confidential, or sensitive information with AI tools whenever possible. This includes information about yourself, other students, staff members or any data that could identify an individual.
- The advice provided in this guide, and in any related University guidance, is intended to support your learning and should be considered guidance rather than a set of mandatory rules. You should always use your own judgement and follow any instructions provided by your module leaders or the University.
- AI-generated content is not always accurate. AI tools can sometimes produce incorrect, misleading, incomplete, or fabricated information (often referred to as "hallucinations"). You should carefully review, verify and critically evaluate all AI-generated outputs before using them in your studies.
- Use AI as a learning aid, not as a substitute for your own work. AI can help you understand concepts, organise ideas, practise language skills and receive feedback, but it should not be used to plagiarise or misrepresent work as your own. Always follow Cardiff University's policies on academic integrity and appropriate use of AI.

Google Gemini

What discords can this tool help with?

ADHD

This guide is informed by a comparison of Google Gemini capabilities and the challenges that learners with ADHD may experience in academic settings. The sections below explain how the Google Gemini tasks included throughout this guide can support learners with ADHD in managing their studies more effectively.

From our research, we have compared tasks that can be completed with Google Gemini, and its characteristics to issues students with ADHD may experience. Below will describe how the following tasks we have set out further along with this guide, involving Google Gemini, can help learners with ADHD.

Low-stakes conversations

Building up conversational skills in a foreign language can be simplified with Google Gemini. You can have realistic conversations with Google Gemini. This eliminates the possible performance anxiety of speaking a new language in front of other people, while still gaining valuable practise. This makes practising conversational skills easier and more accessible, along with helping eliminate the mental resilience possibly needed to practise this skill in other situations. Research suggests that that eliminating tasks that require mental resilience and unnecessary commitment may be beneficial for a learner with ADHD.

Dyslexia

This guide is informed by a comparison of Google Gemini capabilities and the challenges that learners with dyslexia may experience in academic settings. The sections below explain how the Google Gemini tasks included throughout this guide can support learners with dyslexia in managing their studies more effectively.

Based on the findings of this research, a comparison was undertaken between the functionality of Google Gemini and challenges frequently experienced by students with ADHD. The following sections describe how the tasks and activities presented in this guide may help support learners with ADHD throughout their academic journey. Below will describe how the following tasks we have set out further along with this guide, involving Google Gemini, can help learners with dyslexia.

Audio playback for worksheets

Google Gemini can read back learning materials with a realistic voice output. We have identified that students with dyslexia may benefit from being able to have audio outputs rather than text-based outputs, due to the possible difficulty with reading that they may

experience. This allows for learners that have trouble reading to complete tasks more efficiently, by possibly spending less time trying to digest the information presented.

Low-stakes conversations

This feature of Google Gemini allows neurodivergent students to practise realistic verbal language skills, independently. Research suggests that learners with dyslexia may have more success from auditory learning. Due to this, being able to have low-stakes conversations can be majorly beneficial to learners with dyslexia, as this may help them develop language skills independently, without mainly requiring reading.

Processing disorders

This guide is informed by a comparison of Google Gemini's capabilities and the challenges that learners with processing disorders may experience in academic settings. The sections below explain how the Google Gemini tasks included throughout this guide can support learners with processing disorders in managing their studies more effectively.

Based on the findings of this research with Google Gemini, and its characteristics to issues students with processing disorders may experience. Below will describe how the following tasks we have set out further along with this guide, involving Google Gemini, can help learners with processing disorders.

Getting transcripts

Learners with processing disorders may have a harder time processing auditory information at the same speed as others. Google Gemini can help by creating accurate transcripts of audio and video files. A learner can upload recorded lectures, or other audio and video files, and get a full transcription in minutes. Research on this field reveals that this may help a learner that struggled with auditory information processing and will help them learn at their own pace.

Google Gemini will also transcript a live conversation. For example, asking it to transcribe audio live is possible when you're in live mode. This means that for example, a learner can also transcript a conversation or lesson in real time.

Modern languages assessments/ tasks this tool can help with

Oral Examinations

Structured spoken assessments evaluating fluency, pronunciation, interaction and communicative competence.

Example prompt

"Act as a Spanish oral examiner [Role]. Ask me one follow-up question at a time about environmental issues and provide brief feedback on my answers [Task]. I am preparing for a structured university oral examination [Context]."

Interviews/vivas

Simulate interview scenarios.

Example prompt

"Act as an academic panel interviewer [Role]. interview me about my year abroad experience along with 3 likely follow-up questions [Task]. I am preparing for an interactive viva assessment for my language degree [Context]."

Presentations

Individual or group presentations delivered in the target language and/or English.

Example Prompt

"Act as an audience member of a presentation [Role]. I am going to present a slide deck on French identity and multiculturalism.

Desktop

Setting up/ Signing in

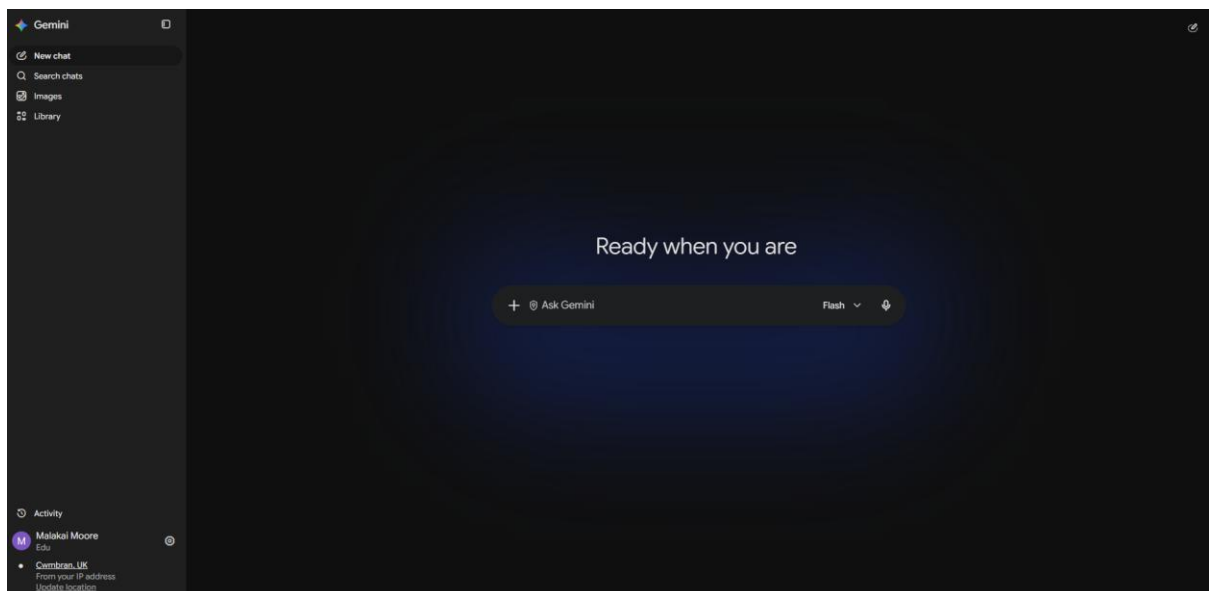
How to do this

You can access Google Gemini but simply searching “Google Gemini” into Google Chrome and select the first appropriate result. From that point you can sign in with your Cardiff University email address.

Disclaimer: We suggest that only sign in/ sign up to the AI tools we suggest with your Cardiff University email address where possible

After set up/ signing in

You should be greeted with the following interface:



From this point, you have access to Microsoft 365 Copilot, provided by Cardiff University. The bar that contains “Ask Gemini” is where you can interact with Google Gemini. And the menu on the left contains (or will contain) your chat history, with the option to create a new chat.

The “Ask Gemini” bar is where you want to type out, your information. *It is worth noting that Copilot allows you to use voice inputs, with the microphone option to the right most part of the “Ask Gemini” bar*

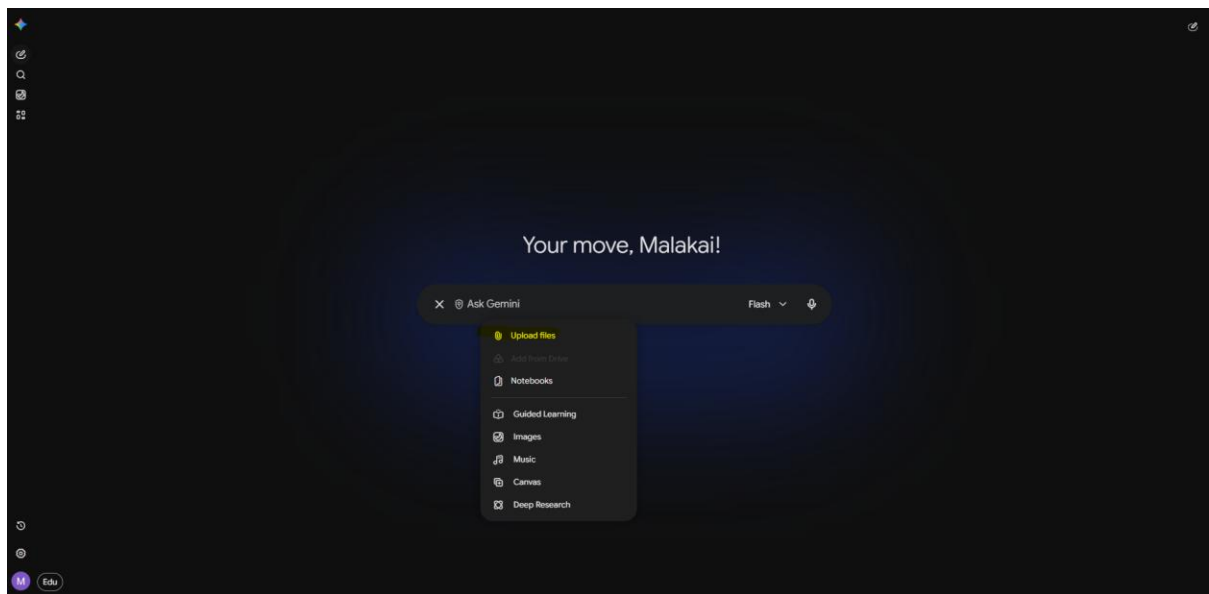
Getting transcripts

To showcase how this can be done, a video produced by Cardiff University Engineering will be uploaded to Google Gemini and then transcribed.

This will be uploaded as a .mp4 file. You can upload files by clicking the “+” symbol that is to the left most part of the “Ask Gemini” bar and then clicking “Upload files”. At which point will bring up your file explorer. Or you can simply drag and drop files from your file explorer.

Disclaimer: You can upload .mp3 and .mp4 files to get transcripts

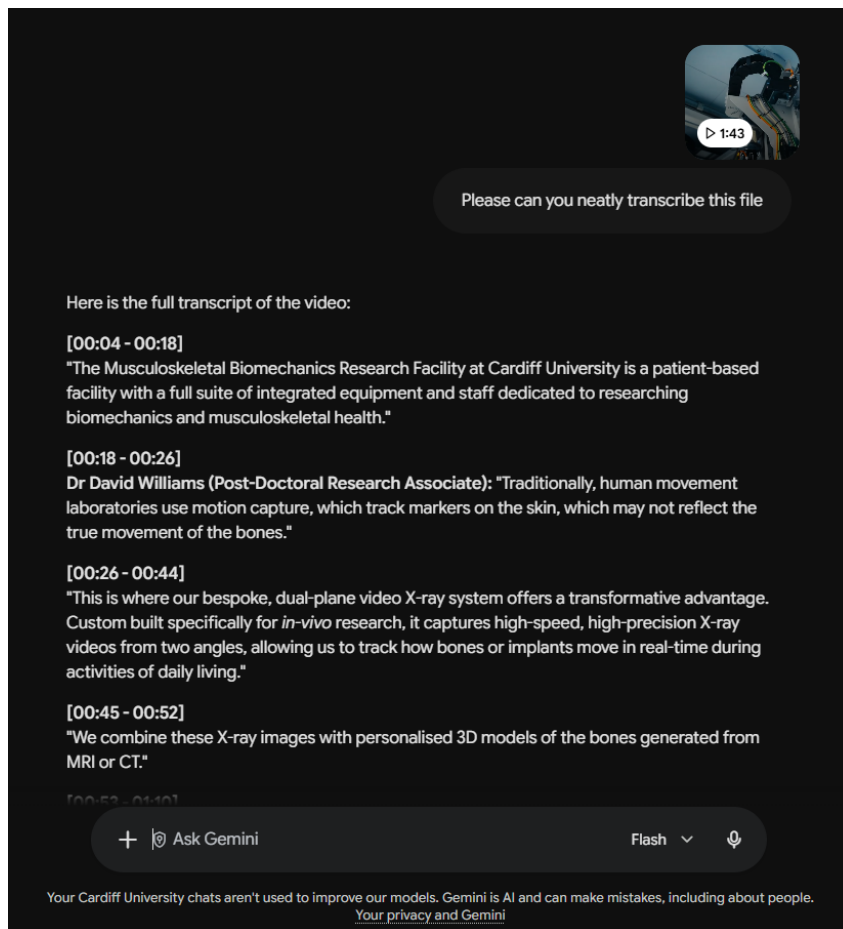
For reference:



An example of a prompt you could give alongside uploading an .mp3 file, or .mp4 file could be as follows:

“Please can you neatly transcribe this file”

Here is the result of this prompt being used alongside a .mp4 files, published by Cardiff University Engineering:



Google Gemini was able to neatly transcribe the .mp4 file in full, perfectly performing the task.

You can apply this tool to lectures. Recording a lecture and then asking for Gemini to transcribe it can be a helpful way to learn. On top of this, being that Gemini is similar Copilot, you can customise the transcript, to view it however you like. For instance, you can ask Gemini to summarise it and change the format.

Mobile

Setting up/ Signing in

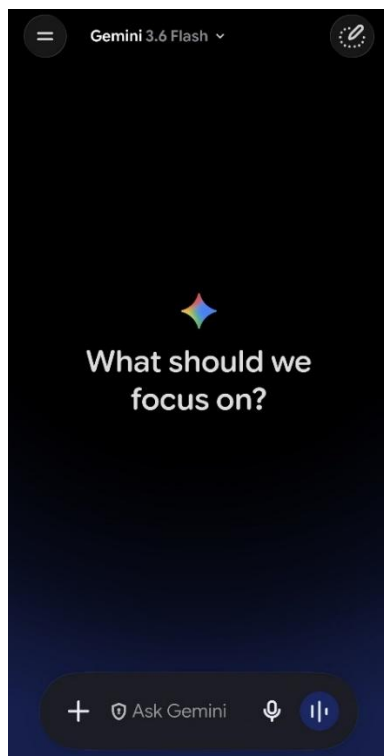
How to do this

You can access Google Gemini by simply going to your mobile's app store and search "Google Gemini" and install "Google Gemini". From that point you can sign in with your Cardiff University email address.

Disclaimer: We suggest that only sign in/ sign up to the AI tools we suggest with your Cardiff University email address where possible

After set up/ signing in

You should be greeted with the following interface:



From this point, you have access to Google Gemini, provided by Cardiff University. The bar that contains "Ask Gemini" is where you can interact with Gemini. The button on the top left will show a side menu that contains (or will contain) your chat history, with the option to create a new chat.

The "Ask Gemini" bar is where you can type your information. *It is worth noting that Gemini allows you to use voice inputs, with the microphone option to the right most part of the "Ask Gemini" bar.*

Asking for audio playback with worksheets

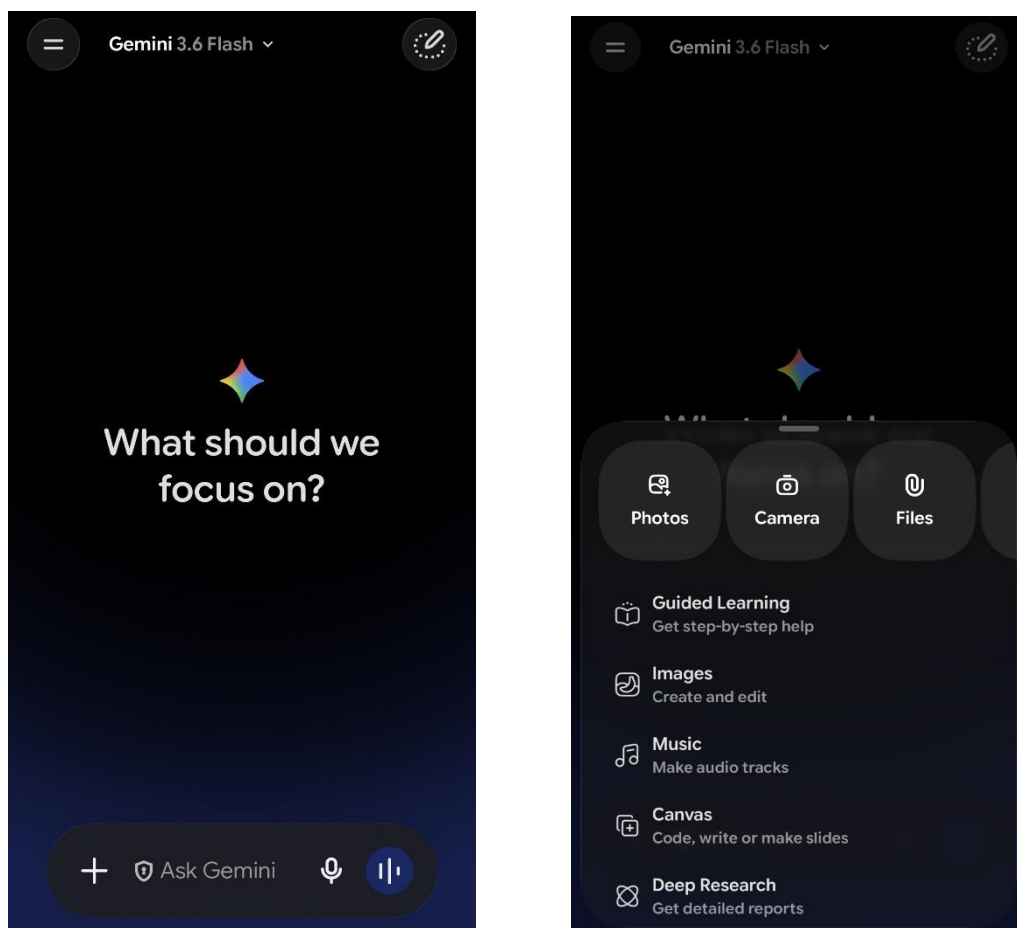
Google Gemini can help you by reading back to you. Outputs from Google Gemini allow it to be read back to. Requesting Gemini to do this will give you back a text-based response, with a playback option, that its voice model will read to you.

How to do this

To demonstrate how this works, an image of a Spanish text will be uploaded to Gemini, asking it to read back a text. Gemini will deliver a response that is text based but has the option that allows it to be read out loud with the voice model.

Firstly, click the “+” symbol that’s to the far left of the “Ask Gemini” bar. Then click the “Camara button”

For reference:

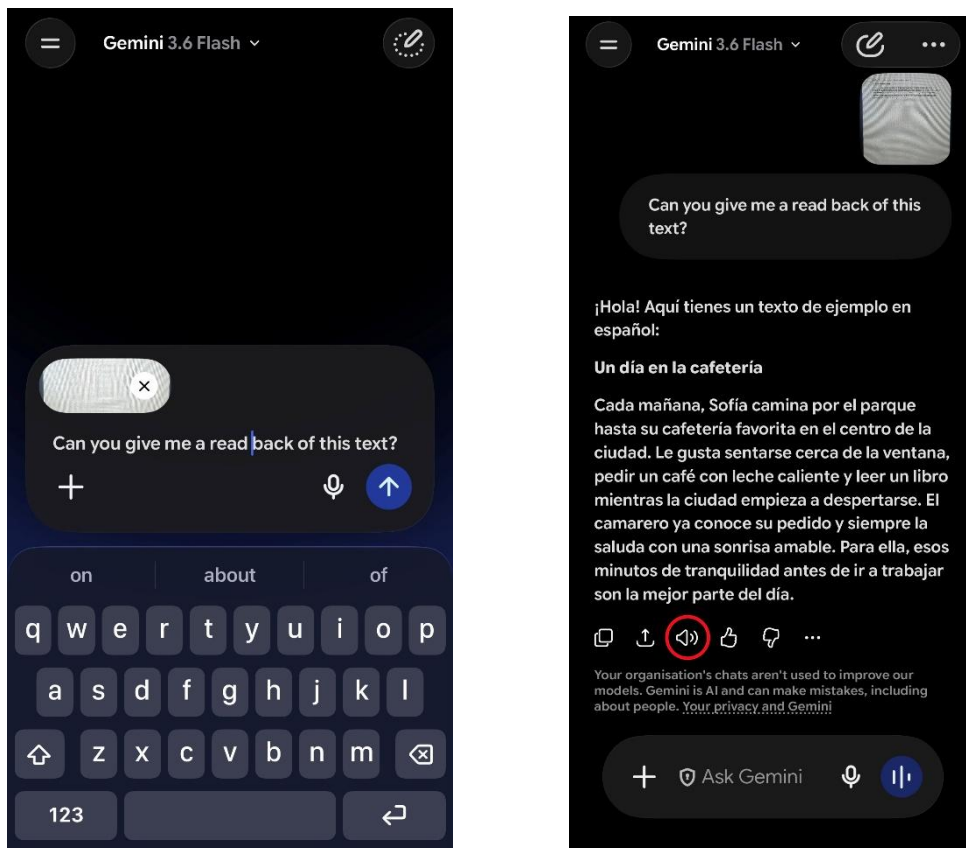


Then take a snap of the text. Once uploaded, pair it with a text-based request. Then at the end of the output, there will be an option to play the response out loud, with the voice model built in.

An example prompt that could be used for this is as follows:

“Can I get a readback of this worksheet?” (with the worksheet attached)

For reference:



This was the text uploaded to Gemini to demonstrate this:

¡Hola! Aquí tienes un texto de ejemplo en español:

Un día en la cafetería

Cada mañana, Sofía camina por el parque hasta su cafetería favorita en el centro de la ciudad. Le gusta sentarse cerca de la ventana, pedir un café con leche caliente y leer un libro mientras la ciudad empieza a despertarse. El camarero ya conoce su pedido y siempre la saluda con una sonrisa amable. Para ella, esos minutos de tranquilidad antes de ir a trabajar son la mejor parte del día.

Gemini was able to successfully transcribe this text. This can be done for worksheets, and other recourses of a similar nature.

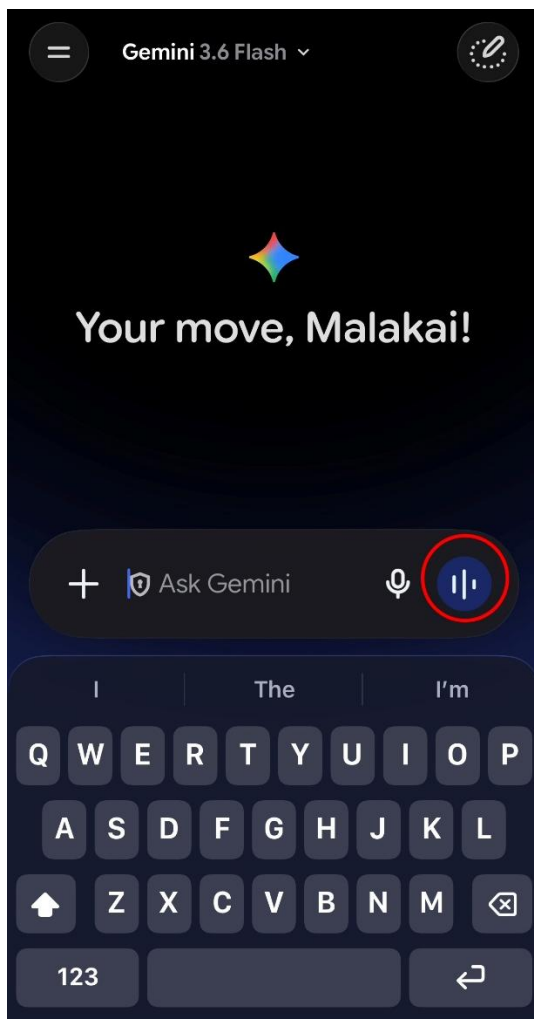
Low-stakes conversations

Google Gemini has a voice conversation model, that allows for realistic conversations to be held, in an array of different languages. This can be used to improve language learning and help learners build confidence speaking a new language.

How do to this

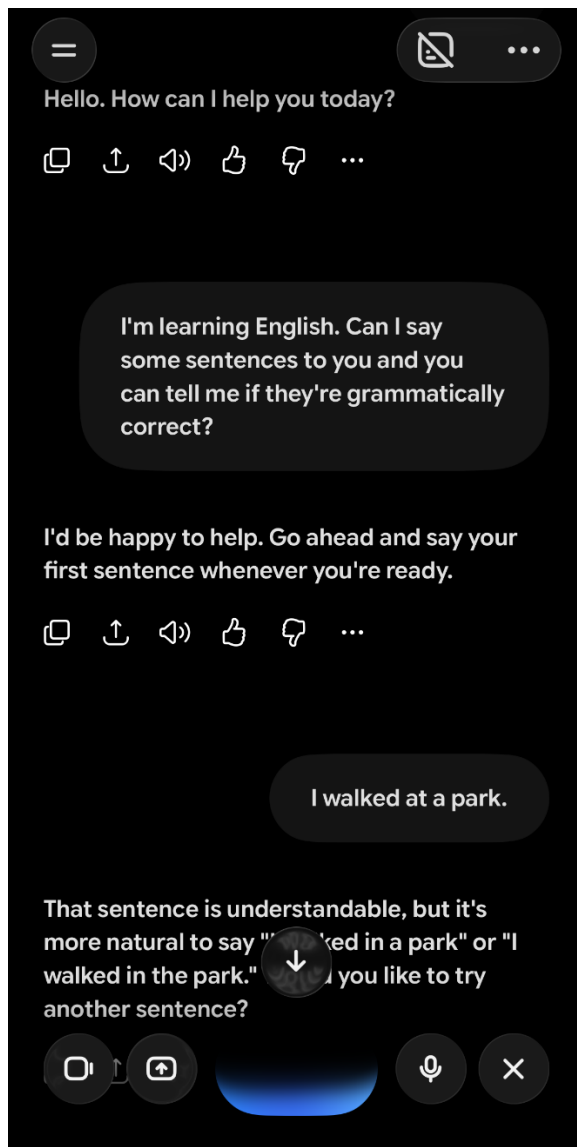
To start this process off, you need to click the voice button, that is to the far right of the “Ask Gemini” bar.

For reference:



From the point of clicking this button, you need to speak to Gemini how you would a human, in the language of your choice. It is that simple.

Here is a small snippet from testing this feature as a person learning English. *The following is a transcript of part of the conversation:*



Google Gemini was able to aid this person learning English, via a realistic conversation

Further recourses (Provided by Cardiff University)

Further guidance for using Google Gemini

<https://intranet.cardiff.ac.uk/students/applications/view/gemini>

References

Child Mind Institute (2026) What is neurodiversity? Available at:

<https://childmind.org/article/what-is-neurodiversity/> (Accessed: 3 September 2026).

Eves, L. (2026) Neuroinclusive teaching in higher education. Higher Education Policy Institute. Available at: <https://www.hepi.ac.uk/2026/01/26/neuroinclusive-teaching-in-higher-education/> (Accessed: 3 September 2026).

For Dyslexia

University of Strathclyde (n.d.) Barriers faced by students with dyslexia. Available at: <https://www.strath.ac.uk/professionalservices/disabilityandwellbeing/informationforstaff/workingwithstudentswithdyslexia/barriersfacedbystudentswithdyslexia/> (Accessed: 2 July 2026).

For Autism

Reachout ASC (n.d.) Autism and behaviour in secondary school. Available at:

<https://reachoutasc.com/autism-and-behaviour-in-secondary-school/>

(Accessed: 2 July 2026).

For processing disorders

National Center for Learning Disabilities (n.d.) Visual and auditory processing disorders.

Available at: <https://www.ldonline.org/ld-topics/processing-deficits/visual-and-auditory-processing-disorders> (Accessed: 10 September 2026)

For Dyspraxia

Dyspraxia Foundation (2012) Dyspraxia in further and higher education. Available at:

<https://dyspraxiafoundation.co.uk/wp-content/uploads/2014/12/Dyspraxia-in-Further-and-Higher-Education-revised-2012.pdf> (Accessed: 10 September 2026).

For ADHD

Royal Borough of Windsor and Maidenhead (n.d.) Barriers to learning: poor attention and concentration and/or hyperactivity (ADHD). AfC Info. Available at:

[https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-](https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-)

[advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-](https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-)

to-learning/barrier-to-learning-poor-attention-and-concentration-and-or-hyperactivity-adhd (Accessed: 10 September 2026).

Cardiff University Support services

<https://www.cardiff.ac.uk/study/student-life/student-support/disability-and-dyslexia-service>